PROJECT OVERVIEW

COVID-19 Trends & Forecasting Analysis

Eko Hospital

Objective: Analyze a 1,000-day COVID-19 dataset covering cases, deaths, recoveries, and testing to identify pandemic trends for public-health monitoring; calculate mortality and recovery rates, examine the relationship between testing volume and case detection, and forecast cumulative cases using an ARIMA time-series model.

Client	Eko Hospital
Dataset	1,000 days of COVID-19 observations
Analysis focus	Pandemic trends, testing relationships, rates, and case forecasting
Key methods	Data analysis, correlation analysis, time-series analysis, ARIMA, visualization
Core variables	Cases, deaths, recoveries, testing
Key outputs	Trend insights, mortality/recovery rates, testing-case relationship, cumulative-case forecast

Executive Summary

Eko Hospital's COVID-19 analysis uses a 1,000-day dataset to examine how cases, deaths, recoveries, and testing change over time. The project is structured around practical public-health surveillance questions: whether infections are increasing or decreasing, how testing volume relates to case detection, how mortality and recovery rates evolve, and whether future cumulative cases can be forecast.

The analytical workflow combines data exploration and cleaning, trend analysis, rate calculations, correlation analysis, time-series forecasting, and data visualization. An ARIMA model is used to forecast cumulative cases, while the relationship between testing volume and case detection is examined through correlation analysis.

The resulting analysis translates the dataset into visual and trend-based insights intended to support outbreak monitoring, public-health surveillance, preparedness, and data-driven resource-planning decisions.

1. Business Context & Analytical Objective

During the COVID-19 pandemic, hospitals and public-health organizations needed timely information to understand infection patterns, monitor severity, evaluate testing activity, and prepare for potential future waves. For Eko Hospital, the project focuses on transforming historical COVID-19 data into clear evidence for surveillance, outbreak monitoring, and resource planning.

The dataset includes new and cumulative cases, deaths, recoveries, and tests, providing a basis for examining infection trends, severity, recovery patterns, and testing activity.

Analytical Objectives

- Prepare and explore the 1,000-day COVID-19 dataset for reliable analysis.
- Analyze daily and cumulative trends in cases, deaths, recoveries, and testing.
- Calculate and visualize mortality and recovery rates over time.
- Examine the relationship between testing volume and case detection using correlation analysis.
- Forecast cumulative COVID-19 cases using an ARIMA time-series model.
- Translate analytical results into visual and trend-based insights for public-health monitoring and planning.

2. Data Preparation & Feature Engineering

The project begins by inspecting the dataset for missing values, outliers, and inconsistencies before trend and forecasting work. Numerical COVID-19 measures are organized by day to support descriptive analysis, rate calculations, correlation analysis, and time-series modelling.

Feature group	Examples	Purpose
Cases	New and cumulative cases	Track infection spread and cumulative burden
Deaths	New and cumulative deaths	Measure outbreak severity
Recoveries	New and cumulative recoveries	Assess recovery patterns
Testing	New and cumulative tests	Assess testing activity and case detection
Time	Daily observations over 1,000 days	Support trend and time-series analysis

3. Analytical & Forecasting Approach

After data preparation, the analysis combines descriptive trend analysis, rate calculations, correlation analysis, and time-series forecasting. These methods provide complementary views of pandemic activity and help turn raw observations into interpretable public-health signals.

Analytical Framework

Analytical component	What it helps assess
Trend analysis	Whether cases, deaths, recoveries, and testing are rising or declining over time
Mortality rate	The observed relationship between deaths and COVID-19 cases over time
Recovery rate	Changes in observed recovery patterns over the analysis period
Correlation analysis	The strength and direction of the relationship between testing volume and case detection
ARIMA forecasting	Potential future movement in cumulative cases based on historical time-series patterns
Visualization	Clear communication of trends and analytical findings

4. Results & Insight Areas

The project produces mortality and recovery rates, an examination of the testing-volume/case-detection relationship through correlation analysis, and an ARIMA forecast of cumulative cases. These outputs turn raw observations into interpretable public-health signals.

What the Analysis Reveals

- COVID-19 activity can be monitored through changes in new and cumulative cases over the 1,000-day period.
- Mortality and recovery rates provide additional context beyond raw case counts.
- Testing volume can be examined alongside case detection to identify statistical association in the observed data; correlation does not by itself establish causation.
- ARIMA provides a structured time-series approach for forecasting cumulative cases from historical patterns.
- Visualizations make changes, peaks, declines, and relationships easier to communicate to decision-makers.

The supplied project description does not specify numerical correlation coefficients, mortality/recovery-rate values, ARIMA parameters, forecast values, or model-accuracy metrics; therefore no unsupported figures are added.

5. Business Interpretation

For Eko Hospital, the value of the analysis is its ability to organize a large historical COVID-19 dataset into a structured monitoring framework. By combining trend measures, rates, correlation analysis, forecasting, and visualization, the project provides a broader view of pandemic activity than relying on a single metric.

6. Impact for Eko Hospital

The project provides a data-analysis framework that can support surveillance and planning activities. Its intended value is to help hospital and public-health stakeholders interpret historical trends, monitor potential changes in outbreak activity, understand testing and case-detection patterns, and use forecasts as one input into preparedness and resource-planning decisions.

Delivered Value

- Public-health surveillance: Structured 1,000 days of COVID-19 observations into analyzable trends.
- Trend monitoring: Examined cases, deaths, recoveries, and testing across time.
- Rate analysis: Calculated mortality and recovery rates to add context to case counts.
- Testing insight: Used correlation analysis to examine testing volume alongside case detection.
- Forecasting: Applied ARIMA to cumulative cases for forward-looking trend analysis.
- Decision support: Converted analytical outputs into visual and trend-based insights for monitoring and resource planning.

7. End-to-End Project Workflow

1	Understand the problem	Define COVID-19 surveillance, monitoring, and forecasting objectives.
2	Prepare the data	Inspect, clean, and organize the 1,000-day dataset.
3	Analyze trends	Examine daily and cumulative cases, deaths, recoveries, and testing.
4	Calculate rates	Assess mortality and recovery patterns over time.
5	Analyze correlation	Examine the relationship between testing volume and case detection.
6	Forecast	Apply ARIMA to cumulative case time-series data.
7	Visualize & interpret	Communicate trends and forecasting insights for surveillance and planning.

Conclusion

The COVID-19 Trends & Forecasting Analysis uses a 1,000-day dataset to examine pandemic patterns across cases, deaths, recoveries, and testing. The workflow combines data preparation, trend analysis, mortality and recovery-rate calculations, correlation analysis, ARIMA time-series forecasting, and visualization. The resulting framework is designed to support public-health surveillance, outbreak monitoring, preparedness, and data-driven resource planning.

Project positioning: Data Analysis • Public-Health Analytics • Time-Series Forecasting • ARIMA • Correlation Analysis • Data Visualization • Trend Analysis • Python • Pandas • Matplotlib • Seaborn

Key Skills: Python, Pandas, Matplotlib, Seaborn, ARIMA, Time-Series Forecasting, Correlation Analysis, Data Visualization, Data Analysis